

Question #: 1

A 45-year-old obese factory worker comes to the physician because of a painful swelling in his groin. He explains that it suddenly appeared three weeks ago after he lifted a heavy box from the floor. During surgical repair, the surgeon identifies an artery located lateral to the herniated bowel. This artery anastomoses with which of the following arteries at the posterior surface of the anterior abdominal wall?

- A. Deep circumflex iliac
 - B. Superior epigastric
 - C. Inferior epigastric
 - D. Gonadal
 - E. Superficial circumflex iliac
-

Question #: 2

A 39-year-old primigravida comes to physician because of 3-week history of involuntary leakage of urine, when her intra-abdominal pressure increases during coughing, laughing, or lifting. Diagnosis of stress urinary incontinence is made. Which muscle is most likely affected to cause this condition?

- A. Coccygeus
 - B. Obturator internus
 - C. Pubococcygeus
 - D. Piriformis
 - E. Psoas major
-

Question #: 3

A 27-year-old man comes to the physician because of a 4-day history of persistent coughing and productive yellow sputum. He has a history of frequent wheezing and dysphagia from childhood. A CT scan shows the right subclavian artery forming a vascular ring with the aorta, around the trachea and esophagus. Which of the following is most likely the embryonic mechanism behind the underlying pathology?

- A. Persistence of the distal part of right dorsal aorta
- B. Persistence of the right dorsal aorta with involution of the distal part of left dorsal aorta
- C. Abnormal involution of the right 4th aortic arch artery and proximal right dorsal aorta
- D. Constriction of the aortic arch distal to the ductus arteriosus

Question #: 4

A 39-year-old woman presents to her primary care physician with severe back pain that began after lifting her 2-year-old daughter. The physician suspects a herniated disc and orders an MRI which is shown. Which of the following nerves would most likely be compressed in this injury?

- A. L2
- B. L3
- C. L4
- D. L5
- E. S1

Attachment:



For question 162748.png

Question #: 5

A 63-year-old man comes to the physician because of a 6-month history of low back pain that has not been relieved by pain medication. Physical examination shows tenderness to palpation in the lumbosacral region. The results from the imaging studies is shown in the attached image. Which of the following is the most likely diagnosis?

- A. Spina bifida cystica
- B. Lumbar disc-herniation
- C. Spondylolysis
- D. Spondylolisthesis
- E. Spondylolytic spondylolisthesis

Attachment:



FOR QUESTION 162754.jpg

Question #: 6

A 23-year-old woman is brought to the emergency department because of a 4-hour history of severe pain after forceful eversion of her ankle. Physical examination shows a swollen, tender ankle joint with a limited range of motion. An x-ray of the ankle shows no fractures. MRI of the ankle joint shows the deltoid ligament has been torn from the medial malleolus. In the attached image, which of the following labeled areas most likely corresponds to the injured bony structure?

- A. A
- B. B
- C. C
- D. D
- E. E

Attachment:



Ankle.jpg

Question #: 7

A 10-year-old boy is brought to the emergency department by his parents because of a 2-hour history of an injury to his left foot. Physical examination shows the wound is on the medial aspect of the sole of his foot, just distal to the calcaneus. Damage to the medial plantar nerve is suspected. Which of the following actions would most likely be lost if there is damage to this nerve?

- A. Abduction of the great toe
 - B. Flexion of the great toe
 - C. Adduction of the great toe
 - D. Flexion of all of the toes
 - E. Spreading and bringing together all of the toes
-

Question #: 8

A 24-year-old man comes to the emergency department because of a 3-hour history of a stab wound just above and posterior to the medial malleolus. Physical examination shows loss of sensation on the entire sole of the foot and inability to spread the toes and bring them back together. He also has weakness of flexion of the toes. Which of the following nerves is most likely damaged?

- A. Superficial fibular
 - B. Lateral plantar
 - C. Deep fibular
 - D. Tibial
 - E. Medial plantar
-

Question #: 9

A carpenter suffered a deep cut to the tip of his thumb. Initially the redness, swelling and pain were limited to the injured part of the thumb, but later the entire thumb and thenar eminence became inflamed. Which group of lymph nodes is the first to receive drainage from this injury?

- A. Posterior axillary
 - B. Apical axillary
 - C. Lateral axillary
 - D. Anterior axillary
 - E. Central axillary
-

Question #: 10

A 40-year-old man is brought to the emergency department after he was involved in a motor vehicle collision. Physical examination shows severely weakened flexion of the forearm and loss of sensation on the lateral surface of the forearm. Imaging studies show multiple fractures of the humerus. Which of the following additional findings will most likely be present in this patient?

- A. Weakness in adduction of the wrist
 - B. Weakness in supination of the forearm
 - C. Difficulty in pronation of the forearm
 - D. Loss of sensation on the medial surface of the arm
 - E. Difficulty in extension of the forearm
-

Question #: 11

A 52-year-old woman comes to the physician because of a 6-month history of a lump in her left breast. Physical examination shows inversion of the left nipple, while the right is normal. A mammogram shows a 6-cm mass in the lower lateral quadrant. Which of the following structures is most likely to be affected, leading to the changes seen in this patient's nipple?

- A. Branches of cutaneous pectoral nerves
 - B. Lactiferous ducts
 - C. Mammary arteries and veins
 - D. Mammary lymphatics
 - E. Suspensory (cooper) ligaments
-

Question #: 12

A 68-year-old man comes to the physician because of a 4-week history of numbness and paresthesia in the medial aspect of his left leg. Six weeks ago, he underwent a venous stripping procedure in which the great saphenous vein and its branches were removed. Which of the following nerves is most likely damaged?

- A. Saphenous
 - B. Femoral
 - C. Superficial fibular
 - D. Sural
 - E. Deep fibular
-

Question #: 13

A 30-year-old man comes to the emergency department because of a 2-hour history of a painful

and unstable right knee. Physical examination shows the findings in the photograph. Which of the following structures is most likely damaged?

- A. Anterior cruciate ligament
- B. Medial collateral ligament
- C. Posterior cruciate ligament
- D. Lateral collateral ligament
- E. Patellar tendon

Attachment:



Picture3.png

Question #: 14

A 49-year-old woman comes to the physician because of a 6-month history of progressive bone pain and multiple fractures after minimal trauma. She has end-stage renal disease and has been receiving maintenance hemodialysis for the past 2 years. Physical examination shows tenderness over long bones. Laboratory studies are shown in the table. Which of the following best explains the cause of these findings?

- A. Hypoparathyroidism
- B. Decreased activity of 1-hydroxylase
- C. Reduced formation of cholecalciferol from 7-dehydrocholesterol
- D. Decreased exposure to sunlight
- E. Decreased activity of 25-hydroxylase
- F. Parathormone (PTH) secreting tumor

Attachment:

Parameter	Patient	Reference range
Serum calcium	8.0	8.4-10.2 mg/dL
Serum phosphate	2.5	3-4.5 mg/dL
Serum alkaline phosphatase (ALP)	350	20-70 U/ L

84436.jpg

Question #: 15

A 24-year-old man comes to the physician for a routine health examination. He feels well and has no significant medical history. He does not take any medications or supplements. Physical examination shows no abnormalities. Laboratory studies are shown in the table. Which of the following is the most likely cause of these laboratory findings?

- A. Acute liver damage
- B. Myocardial infarction
- C. Enzyme induction due to alcohol
- D. Bone disease
- E. Acute pancreatitis
- F. Muscular dystrophy

Attachment:

Alkaline phosphatase	Normal
CK-MM	Normal
CK-MB	Normal
γ -glutamyl transpeptidase (GGT)	Elevated
α -Amylase	Normal
Pancreatic lipase	Normal
Alanine aminotransferase	Normal

140465.jpg

Question #: 16

An investigator is studying the properties of a replicative DNA polymerase isolated from a newly discovered bacterial strain. In vitro, studies show that the enzyme is capable of synthesizing DNA in the 5'→3' direction but lacks 3'→5' exonuclease activity. Which of the following processes is most likely defective in this bacterial strain?

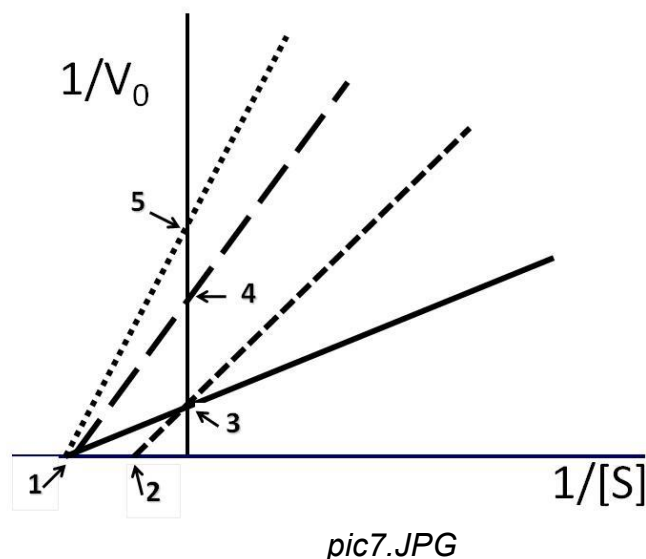
- A. mRNA synthesis
- B. Proof-reading during replication
- C. Splicing and capping of mRNA
- D. Translation and protein synthesis
- E. Unwinding of double-stranded DNA

Question #: 17

An investigator is studying the enzyme kinetics of HMG-CoA reductase using a Lineweaver–Burk plot. The enzyme is analyzed under four conditions: without inhibitor (control), with statin, and with two other unknown inhibitors. Statin is a competitive inhibitor that is a structural analog of HMG-CoA and binds noncovalently to the enzyme's active site. The resulting Lineweaver–Burk plot is shown, with X- and Y-intercepts labeled 1–5. Which of the following points is best used to calculate the V_{max} for the reaction in the presence of statin?

- A. Point 1
- B. Point 2
- C. Point 3
- D. Point 4
- E. Point 5

Attachment:



Question #: 18

A 20-year-old woman comes to the physician because of a 2-week history of gum bleeding while brushing her teeth. She has also noticed easy bruises on her arms and legs. She recently began a strict low-carbohydrate diet consisting mainly of meat, eggs, and cheese. She avoids fresh fruits and vegetables and does not take any vitamin supplements. Physical examination shows swollen, bleeding gums and small petechiae on the forearms. Which of the following functions is most likely defective in this patient?

- A. Activity of lysyl oxidase
 - B. Activity of prolyl hydroxylase
 - C. Synthesis of fibrillin
 - D. Intestinal absorption of calcium and phosphate
 - E. Glycosylation of N-linked glycoproteins
 - F. Assembly of proteoglycans and glycosaminoglycans
-

Question #: 19

A newborn male born at 28 weeks' gestation develops respiratory distress and cyanosis shortly after birth. He is intubated and admitted to the neonatal intensive care unit. A diagnosis of neonatal respiratory distress syndrome is made, and he receives exogenous surfactant replacement therapy containing artificial lung surfactant. Which of the following lipids is the most abundant component of the surfactant used in this patient?

- A. Phosphatidylinositol containing arachidonic acid
 - B. Cholesterol esters containing stearic acid
 - C. Phosphatidylcholine with two saturated fatty acids
 - D. Sphingomyelin containing palmitate
 - E. Dipalmitoyl-phosphatidylethanolamine
-

Question #: 20

A 62-year-old man comes to the physician because of a 3-month history of needing frequent changes in his eyeglass prescription. He also reports increased thirst and urination. Laboratory studies show a fasting plasma glucose level of 180 mg/dL. A diagnosis of type 2 diabetes mellitus is made. Further studies indicate that his hyperglycemia results from decreased glucose uptake by skeletal muscle and adipose tissue due to insulin resistance. The glucose transporters responsible for uptake in these tissues normally mediate which of the following types of transport?

- A. Primary active transport

- B. Secondary active transport
 - C. Co-transport with Na⁺ ions
 - D. Facilitated diffusion
 - E. Antiport with K⁺ ions
-

Question #: 21

An investigator is studying the molecular mechanisms involved in polyglutamine diseases, such as Huntington disease. Analysis of affected cells shows aberrant transcriptional regulation associated with defective acetylation of histones. Which of the following best describes the relationship between histone acetylation and gene transcription?

- A. Deacetylation of histones forms euchromatin
 - B. Deacetylation of histones reduces transcriptional activity
 - C. Acetylation increases the affinity between histones and DNA
 - D. Acetylation causes the recruitment of DNA methyltransferase
 - E. Acetylation activates and recruits topoisomerase
-

Question #: 22

A 2-year-old boy is brought to the physician by his mother because of repetitive self-injurious behavior, including biting of his fingers and lips. She reports that he has had profound developmental delays since early infancy. Physical examination shows global developmental delay, hypotonia, and self-mutilating behavior. Laboratory studies show:

- Markedly elevated serum uric acid
- Increased urinary uric acid excretion

Which of the following inherited enzyme deficiencies is the most likely cause of this patient's condition?

- A. Adenosine deaminase
 - B. Purine nucleoside phosphorylase
 - C. Orotate phosphoribosyl transferase
 - D. Hypoxanthine guanine phosphoribosyl transferase
 - E. Xanthine oxidase
 - F. PRPP synthetase
-

Question #: 23

An investigator isolates a mutant strain of *Escherichia coli* that demonstrates defective

termination of transcription. Further analysis reveals a point mutation in a gene encoding a protein required for proper termination of transcription. Which of the following proteins is most likely to have reduced function in this bacterial strain?

- A. DNA A protein
 - B. Rho factor
 - C. Single strand binding proteins (SSB)
 - D. Initiation factor-1 (IF-1)
 - E. Peptidyl transferase
 - F. Sigma factor
-

Question #: 24

A scientist studying transposable element (or mobile) DNA elements such as LINEs and SINEs makes an analogy that these mobile transposons are usually relatively harmless viruses. However, occasionally, a type of these transposons may insert into a critical gene causing disease. Of course, their activity is usually suppressed (by unknown mechanism), otherwise critical genes might be interrupted, and our genomes would be destroyed. What best explains how these DNA elements expanded in the human genome?

- A. LINEs encode reverse transcriptase
 - B. LINEs propagate during meiosis
 - C. Transposons inhibit normal splicing events
 - D. Transposons hijack telomerase
 - E. LINEs are found within the telomere
 - F. LINEs are found within the centromere
-

Question #: 25

A 3-year-old boy has hemolytic anemia. The hemolysis was due to accumulation of alpha-globin chains in erythrocytes. Which of these hemoglobin disorders is most likely in this patient?

- A. Deletion of four alpha-globin genes
 - B. Mutations in both beta-globin genes
 - C. Mutations in both gamma-globin genes
 - D. A glutamate to valine substitution on sixth amino acid of beta-globin
 - E. A glutamate to lysine substitution on sixth amino acid of beta-globin
-

Question #: 26

A 10-year-old girl is brought to the physician by her parents because of a progressive accumulation of discolorations and growths on her skin. Physical examination suggests that the abnormal tissues are benign neurofibromas. Targeted genetic analysis of her *NF1* gene shows that she has a pathogenic mutation which she inherited through her germline. Testing of the parents shows maternal inheritance. The mother was diagnosed only after genetic testing because she was mildly affected. Which of the following genetic events is most likely to occur first during the manifestation of the numerous neoplasms in this patient?

- A. Mutation of the mitochondrial genome
 - B. *NF1* increases apoptosis
 - C. Accumulation of toxic *NF1* RNA
 - D. *NF1* activates a cell cycle checkpoint
 - E. Loss of *NF1* heterozygosity
-

Question #: 27

A 16-year-old boy is brought to the physician by his parents because he is not developing secondary sexual characteristics. Physical examination shows testicular atrophy, gynecomastia, and normal intelligence. Which of the following cytogenetic findings would be consistent with the most likely diagnosis of this patient?

- A. A paracentric inversion on the X chromosome
 - B. An isochromosome X karyotype: 46,X,i(Xq)
 - C. The presence of one Barr body
 - D. An extra copy of chromosome 21
 - E. Maternal uniparental disomy of chromosome 15
-

Question #: 28

A researcher stimulates a cell culture with insulin-like growth factor to investigate how gene expression is altered in response to this hormone. Poly-T conjugated sepharose beads are used to isolate mRNA by hybridization to the poly-A tail. Reverse transcriptase is used to synthesize single strands of DNA using the captured mRNA as template. Which of the following would best describe the DNA strands that are synthesized in this procedure?

- A. They lack exon sequences
 - B. They contain intron and exon sequences
 - C. They are the promoter sequences
 - D. They lack intron sequences
 - E. They contain intron and promoter sequences
-

Question #: 29

A 52-year-old man is brought to a physician by his wife because of a 3-year history of progressively worsening chorea. His wife reports that he has had personality changes, slight loss of memory and gross sleep disturbances that began when he was in his mid-to-late forties. Which of the following best describes the genetic test that would be used to diagnose the disorder in this patient?

- A. G-band karyotype
 - B. Direct enzyme assay
 - C. Sizing of PCR products
 - D. Northern analysis
 - E. Microarray CGH
-

Question #: 30

A 39-year-old woman comes to the physician because of a 2-year-history of progressively worsening muscle weakness and a reduced ability to release her grip after squeezing. For instance, she has difficulty releasing a doorknob after opening the door. Genetic testing by PCR amplification of a repetitive sequence is diagnostic for a specific neurodegenerative disorder. Which of the following best describes the disorder caused by the pathogenic variant in this patient?

- A. Myoclonic epilepsy
 - B. Muscular dystrophy
 - C. Multiple sclerosis
 - D. Myotonic dystrophy
 - E. Huntington disease
 - F. Fragile X syndrome
 - G. Neurofibromatosis
-

Question #: 31

An investigator is studying the telophase stage of mitosis where microtubules pull sister chromatids to opposite ends of the cell poles. Which of the following best describes the portion of the chromosome that attaches to the microtubules to drive this movement?

- A. Euchromatin
- B. Centromere
- C. Telomere
- D. The *Alu* sequence

Question #: 32

A researcher studying non-syndromic congenital heart defects (CHD) finds that the population incidence of this multifactorial developmental defect is approximately 1 in 100 live births. Comparisons are made to estimate recurrence risk depending on the affected relative. Which of the following best explains an aspect of the inheritance of this disorder?

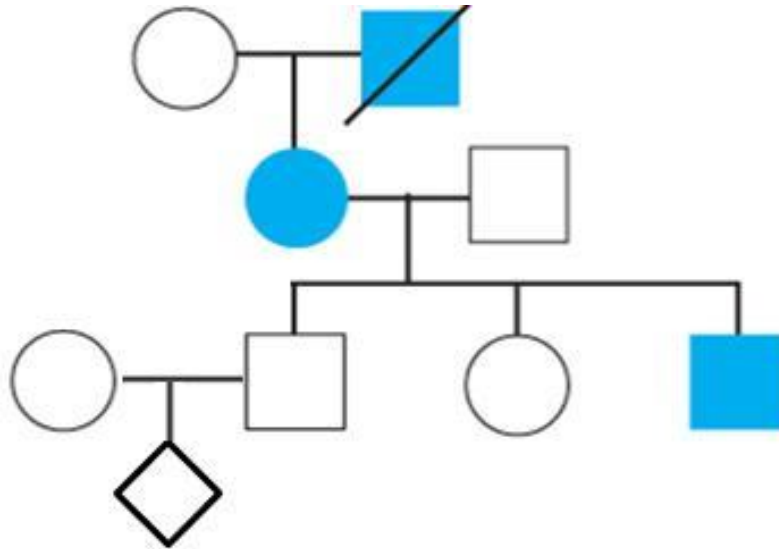
- A. Recurrence risk will be higher in their male children than in their female children
 - B. Recurrence risk reduces dramatically as relatedness decreases
 - C. Recurrence risk will be the same as the population incidence
 - D. Recurrence risk will be one half of the population incidence
 - E. There is no risk to the children as this is a quantitative trait
 - F. Recurrence risk will be twice the population incidence
-

Question #: 33

A scientist is studying a rare genetic disorder to learn more about disease penetrance in people who inherit the disease-causing mutation. A representative pedigree of a family afflicted with the disorder is shown. Which of the following disorders is likely to be in this family?

- A. Prader-Willi syndrome
- B. Duchenne muscular Dystrophy
- C. Neurofibromatosis
- D. Sickle Cell Anemia
- E. Phenylketonuria

Attachment:



ID 8635 _ Exam Soft ID image.JPG

Question #: 34

A 32-year-old woman comes to the physician to learn about the recurrence risk for cystic fibrosis (CF) as she is planning to have a family. She has a niece who is a known carrier for a mutation causative of CF, and her husband is from a population where the incidence of CF is approximately 1 in 2,500. Which of the following is the best recurrence risk estimate of CF for the children of this couple?

- A. 1/50
- B. 1/40
- C. 1/400
- D. 1/2,500
- E. 1/200

Question #: 35

A 63-year-old woman comes to the physician for a routine colonoscopy. During the procedure, a mass is observed in the ascending colon. A biopsy is taken, and a karyotype shows aneuploidy within the cells taken from the mass. The epithelium of the affected structure is usually a single cell layer thick with cells that are taller than they are wide. Which of the following best characterizes this epithelium?

- A. Simple columnar
- B. Transitional
- C. Pseudostratified ciliated

- D. Stratified cuboidal
 - E. Stratified squamous
-

Question #: 36

A 21-year-old woman comes to the physician because of a 6-month history of severe acne on her face. Physical examination shows some pimples are acutely inflamed and a few have some pus inside, which can be easily expressed out. The sebaceous glands involved in this condition have a secretory process wherein the entire cell becomes a part of the secretory product. Which of the following best describes this mode of secretion?

- A. Merocrine
 - B. Holocrine
 - C. Autocrine
 - D. Paracrine
 - E. Apocrine
-

Question #: 37

A 45-year-old man comes to the physician because of a 24-hour history of extensive skin lesions, fever, and dehydration. Physical examination of the skin lesions shows small vesicles which are easy to rupture. He is diagnosed with pemphigus vulgaris, which involves separation of cells within the stratum spinosum layer of the epidermis. Which of the following cell junctions are most likely affected in this condition?

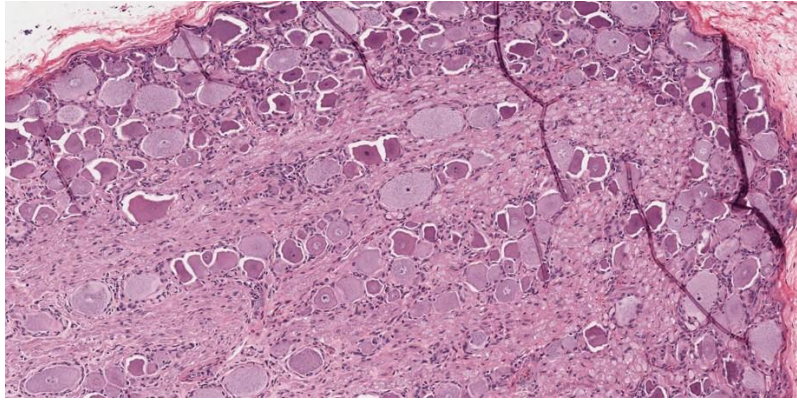
- A. Hemidesmosomes
 - B. Gap junctions
 - C. Zonula occludens
 - D. Zonula adherens
 - E. Desmosomes
-

Question #: 38

A 17-year-old boy is brought to the emergency department because of a puncture wound to the mid back region after falling from a mango tree 30-minutes ago. Physical examination shows a penetrating wound adjacent to the vertebral column. He is admitted to the surgical ward for exploration. During surgery, a piece of tree branch is found lodged close to the spinal column and has damaged the structure indicated in the photomicrograph. The damaged structure is analogous to the nuclei of the CNS, in that it contains the cell bodies of neurons. Which of the following best describes the microscopic features of the damaged structure?

- A. Synapsing occurring
- B. Relatively small cell bodies
- C. Cells with eccentric nucleus
- D. Large cell bodies with thick bundles of nerve fibers
- E. Cells with multiple dendrites and a single axon
- F. Thin, randomly arranged nerve fibers

Attachment:



DRG.jpg

Question #: 39

A 45-year-old man comes to the physician because of a 3-month history of sharp pain in the right buttock and thigh. He is a mechanic. Physical examination shows positive straight leg raise test. MRI shows a herniation of the L4-L5 intervertebral disc. Which of the following is most likely a characteristic feature of the cartilage component of this structure?

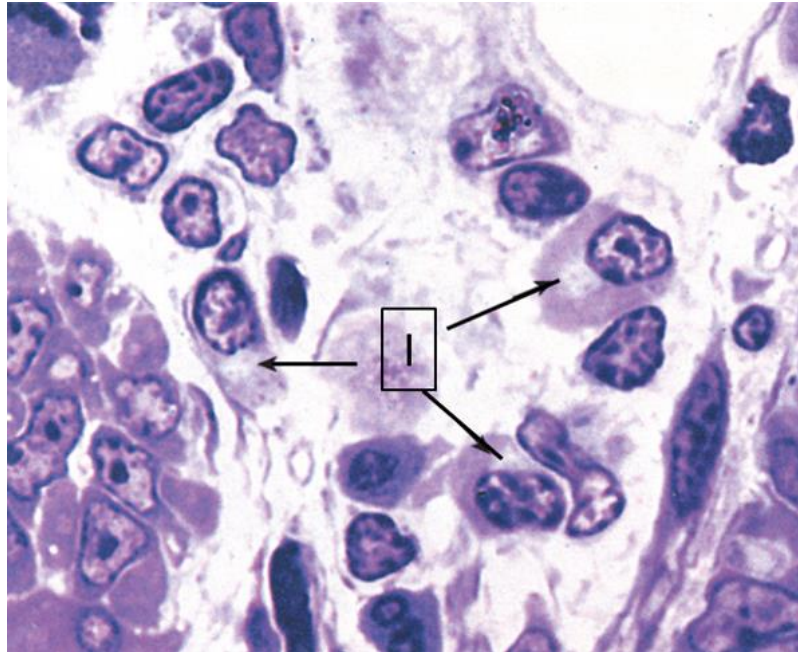
- A. Matrix contains only Type II collagen fibrils
- B. Linear isogenous groups
- C. Matrix contains abundant elastic fibers
- D. Firmly attached perichondrium
- E. High capability for repair
- F. Undergoes both appositional and interstitial growth

Question #: 40

A 20-year-old man comes to the emergency department because of a 2-month history of fever and bone pains. Physical examination shows generalized lymphadenopathy and tenderness of the left tibia with an associated limp. An x-ray shows a mass within the tibia. Laboratory studies of blood show increased antibodies. Biopsy shows a tumor originating from the cells in the attached light micrograph. He is diagnosed with plasmacytoma. Which of the following organelles is most likely indicated by (I) in the light micrograph?

- A. Rough endoplasmic reticulum
- B. Smooth endoplasmic reticulum
- C. Golgi complex
- D. Lysosomes
- E. Mitochondria

Attachment:



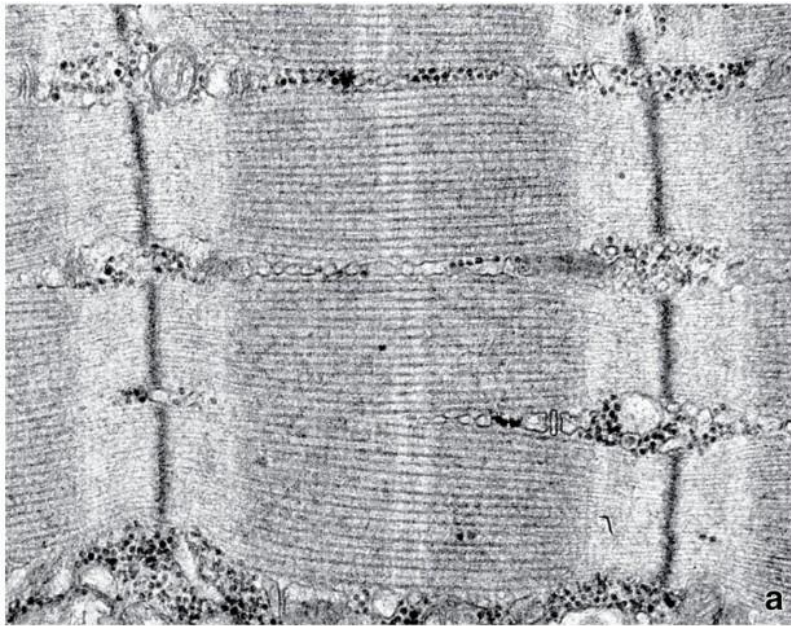
golgi .png

Question #: 41

A 17-year-old boy is brought to the physician by his parents because of a 4-month history of lower limb weakness. Physical examination shows atrophy of the muscles of both lower limbs and difficulty walking. A biopsy is obtained to rule out myopathy and is shown in the attached image. Molecular studies show antigen-antibody complexes against a protein that keeps thick filaments centered between the Z-lines of the sarcomere and prevents excessive stretching. Which of the following proteins is most likely affected?

- A. Tropomyosin
- B. Titin
- C. Actin
- D. Myosin
- E. Desmin
- F. Dystrophin

Attachment:



Sarcomere.jpg

Question #: 42

A 32-year-old man comes to the physician because of a 6-week history of progressively worsening weakened coordination of the arms (ataxia) and problems with speech articulation (dysarthria). He is currently undergoing chemotherapy and radiotherapy for small cell lung cancer. Brain CT scan shows atrophy of the cerebellum. Immunohistochemistry shows antibodies against Purkinje cells. He is diagnosed with paraneoplastic subacute cerebellar degeneration. Which of the following neurons best characterizes the targeted cell that is resulting in the symptoms seen in this patient?

- A. Pseudounipolar
- B. Multipolar
- C. Golgi type II
- D. Bipolar
- E. Unipolar

Question #: 43

A 56-year-old woman is administered an intravenous dose of an antibiotic. Plasma drug concentrations are determined at 2 hours and 4 hours after administration. The drug concentrations are found to be 2 mg/L and 0.5 mg/L, respectively. The antibiotic follows first-order kinetics of elimination. Which of the following is most likely to be the half-life of elimination of the antibiotic?

- A. 0.5 hours
 - B. 1 hour
 - C. 1.5 hours
 - D. 2 hours
 - E. 2.5 hours
-

Question #: 44

A 67-year-old man comes to the physician because of fatigue and lethargy. One month ago, he was diagnosed with stage 1 hypertension and started on propranolol. His pulse is 52/min, compared with 78/min one month ago. The physician switches him to pindolol. His symptoms resolve and his heart rate increases to 69/min. Which of the following properties of pindolol best explains its effect on this patient's heart rate?

- A. It is a competitive antagonist
 - B. It is a full agonist
 - C. It is a partial agonist
 - D. It is an inverse agonist
 - E. It is a noncompetitive antagonist
-

Question #: 45

A 34-year-old man comes to the emergency department because of a 1-hour history of a drug overdose. The physician decides that the patient needs alkalinization of the urine. After the process is complete, there is a significant increase in the renal clearance of the drug. Which of the following pharmacokinetic properties most likely describes the causative drug?

- A. The drug is a strong acid
 - B. The drug is a weak acid
 - C. The drug is a nonelectrolyte
 - D. The drug is a weak base
 - E. The drug is a strong base
-

Question #: 46

A 56-year-old man comes to the emergency department with a 5-year history of myocardial infarction, heart failure, and arrhythmia. He receives lidocaine by continuous IV infusion. The target plasma concentration is 3 mg/L. The pharmacokinetic parameters for lidocaine in the general population are V_d of 70 L, Clearance of 35 L/h, and half-life of 1.4 h. An infusion has begun. The plasma concentration of lidocaine is measured 2.8 h later and reported to be 2.4

mg/L. Which of the following is most likely to be the steady state plasma concentration in this patient?

- A. 1.5 mg/L
 - B. 2.4 mg/L
 - C. 3.2 mg/L
 - D. 4.6 mg/L
 - E. 6.9 mg/L
-

Question #: 47

A 68-year-old woman comes to the physician because of a 2-week history of headaches. She is currently on prescription medication, but she cannot recall the name of the drug she is taking. Her blood pressure is 185/95 mm Hg. Physical examination shows tachycardia. Which of the following types of drugs is most likely to be her prescription medicine?

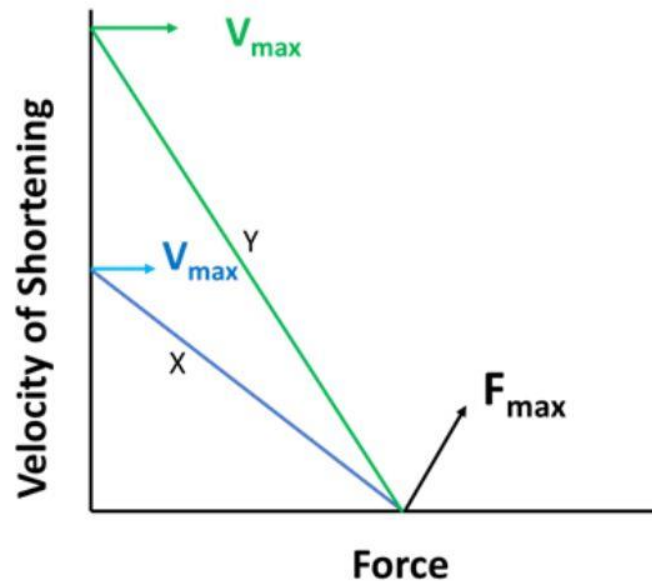
- A. Alpha-1 adrenergic antagonist
 - B. Alpha-2 adrenergic antagonist
 - C. Beta-1 adrenergic agonist
 - D. Beta-2 adrenergic agonist
 - E. Muscarinic agonist
-

Question #: 48

A 30-year-old man comes to the physician for a routine examination. He is an athlete training for a competition, and over the past two months he has been training to improve the contractile force of his leg muscles by lifting a series of increased loads. The attached diagram represents the force velocity relationship from his leg muscles before (curve X) and after (curve Y) the 2 months of training. Which of the following explanations is the most likely reason for the shift from curve X to curve Y?

- A. Decreased myosin ATPase activity
- B. Increased afterload
- C. Increased myosin ATPase activity
- D. Increased preload
- E. Recruitment of larger motor units

Attachment:



Capture.JPG

Question #: 49

A 16-year-old boy is brought to the emergency room after being unable to walk and struggling to stand without help after competing at an athletics event. He says he feels extremely weak. He has a family history of primary hypokalemic periodic paralysis. Laboratory studies show that his plasma potassium level is 2.1 mmol/L (normal range 3.5 - 5 mmol/L). He is treated with a K^+ supplement in a saline IV infusion. Which of the following is the most likely primary cause of his weakness?

- A. Depolarized rest membrane potential of excitable cells
- B. Hyperpolarized resting membrane potential of excitable cells
- C. Increased refractory period of voltage gated sodium channels
- D. Depolarizing blockade of voltage gated sodium channels
- E. Increased opening of delayed rectifier potassium channels

Question #: 50

Nasal epithelial cells are isolated from human nasal brushings and bathed in an extracellular solution containing Na^+ 140 mM, K^+ 5 mM, Cl^- 105 mM. Assume normal intracellular electrolyte concentrations of Na^+ 14 mM, K^+ 140 mM, and Cl^- 40 mM. The membrane potential of one of the epithelial cells is measured and found to be -60 mV. Which of the following combinations best describes the driving force and direction of flux for sodium across the cell membrane?

A	120 mV	influx
B	120 mV	efflux
C	60 mV	influx
D	60 mV	efflux
E	30 mV	influx
F	30 mV	efflux

- A. A
- B. B
- C. C
- D. D
- E. E
- F. F